

Integrated Optical Waveguides in LiNbO₃ : Modelling and Experimental analysis

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Abstract

Graded-index optical channel waveguides are fabricated by diffusion of Ti in LiNbO₃. A model has been developed for calculating the mode profiles of the same, by first finding the effective refractive index profile along the lateral direction, and then discretising the profile and applying the matrix method. This effective-index-based matrix method (EIMM) has been presented in this paper and established as a computationally fast, analytical method to compute modal electric field and intensity profiles in single and coupled graded-index channel waveguides. The method can be used in coupled systems consisting of three or more waveguides where the input power may be launched to any of the waveguides. Some of the computed intensity profiles have been compared with experimental data. Although computation and measurements have been performed only for TE and TM polarisations, the method has the inherent capability of handling arbitrary polarisation.

Keywords : Ti:LiNbO₃, effective-index-based matrix method, mode profile, coupled waveguides

Introduction

Optical waveguides are the basic building blocks in integrated optical devices and optical integrated circuits which play an important role in today's communication systems. They are formed by in-diffusion, ion-exchange, ion-implantation, [1-3] etc. Lithium niobate (LiNbO₃) is the most widely used substrate material for formation of optical waveguides because of its excellent electro-optic and acousto-optic property. Diffusion of transition elements is the commonest procedure to form channel waveguides in LiNbO₃, and Titanium (Ti) as the transition element is so far the best choice. Fabrication of good-quality Ti:LiNbO₃ channel waveguides is not a new challenge, and the related technological problems and their remedies have been well researched and understood [4-5]. However, the mode profiles of the fabricated single and coupled waveguides at any of the propagation distance need to be optimised before fabricating a device, because it suggests the way to modify the index profile that will enable light to be guided in a desired way for use in various applications. Thus the need of a good model for mode-profile calculation becomes important.

For step-index channel waveguides, analytical methods already exist for the computation of mode profiles. For graded-index channel waveguides with complex refractive index profiles, the numerical beam propagation method (BPM) is applicable. The 2D/3D BPM is, however, highly computation intensive. In this paper, the results of detailed analysis of waveguides and directional couplers are presented. In section 2, the fabrication process and structural investigation of Ti:LiNbO₃ waveguides are described. Proper fabrication condition (time, temperature, ambient), and dimensions (width, thickness), material data, initial field distribution, etc. are required for low propagation loss, strict optical confinement in the waveguide and low fiber-waveguide coupling loss, and this can be only achieved by proper fabrication experience. The theoretical model based on the effective-index-based matrix method (EIMM) is presented in section 3 in a very short form. The advantage of this model is that it is comparatively faster than the BPM. The mathematical details of the model are presented elsewhere [7]. The experimental and numerical results obtained at 1.31 μm operating wavelength are compared and discussed in section 4, and the paper is concluded in section 5.

Fabrication and structural investigations

Fabrication

LiNbO₃ wafers were first cleaned properly and then coated with titanium (Ti) of thickness 95 nm on the optically polished Z⁻ surface because this surface is more immune to ferroelectric domain inversion in high-temperature treatment (which strongly influences repeatable device performance) than the Z⁺ surface [8]. The required Ti thickness for single-mode operation of the waveguide at 1.31 μ m transmitting wavelength extends over a range roughly from 600 to 1100 Å [9,10], but the mode-size and the rate of the decaying tails of the mode profile change radically with the metal thickness. These tails and therefore the Ti-thickness strongly influence the inter-waveguide coupling coefficient and the bending loss. Also, the mode size determines the fibre-to-waveguide coupling loss. Thus the tolerance on Ti-thickness of a single-mode Ti:LiNbO₃ waveguide is particularly stringent compared to other fabrication parameters. Different waveguide structures (straight waveguides, couplers, etc) are patterned on the Ti film and are diffused into LiNbO₃ substrate at 1050°C for 6 hr in a flow of wet O₂ ambient which is introduced into the furnace through a bubbler containing DI water. The gas flow rate and bubbler temperature are maintained at 1 lit/min and 60°C, respectively throughout the diffusion process. The amount of hydrogen introduced within the crystal in wet O₂ ambient affects lithium mobility sufficiently to prevent Li₂O out-diffusion and surface guiding of the substrate [11].

Structural Investigations

When Ti metal diffuses into LiNbO₃, there is a swelling at the surface. This surface swelling can be seen when viewed through an optical microscope with reflected light. Figs 1(a) & (b) show topview of the fabricated Ti:LiNbO₃ waveguide in reflected light microscopy where the waveguides (a) and (b) are diffused using water bubbler temperatures 95°C and 60°C, respectively. From these two photomicrographs, it is clear that waveguide surface degrades at higher bubbler temperature. The rms surface roughness of the fabricated waveguides are 35.89 Å and 16.10 Å for Figs.1(a) & (b), respectively, as measured by Dektak-3 surface profiler. It has been reported [12-16] that surface roughening during the heat treatment is due to out-diffusion of Li and epitaxial growth of the LiNb₃O₈ phase on LiNbO₃. However, in more recent experimental work of Nagata and Ichikawa [17], no additional x-ray diffraction peak was detected from visually surface-degraded Ti:LiNbO₃ waveguides, and hence, the origin of the surface precipitates is not yet established. However, a 'critical humid condition' during the diffusion process is necessary to fabricate low-loss Ti:LiNbO₃ waveguide. In the present fabrication process the bubbler temperature is optimized at 60 $\pm$ 5°C to have smooth surface morphology and hence lower propagation loss of the waveguide.

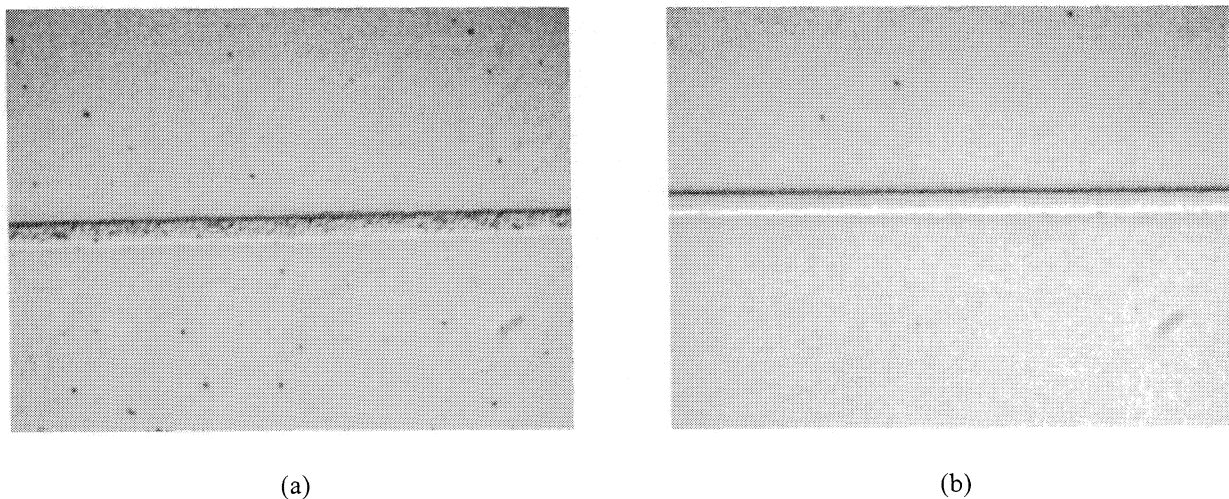


Fig 1. Topview of the fabricated Ti:LiNbO₃ waveguide in reflected light diffused using water bubbler temperatures of (a) 95°C and (b) 60°C

Edge polishing

Light is launched to the waveguides by endfire coupling which requires, for low coupling loss, optically flat waveguide edges free from cracks, scratches, chipping and roll-over. A convenient way to get optically flat waveguide edge is to polish the waveguide by epoxying and pressing a second block of LiNbO₃ against the substrate top surface. Although no rounding of the edge is detectable if the epoxy layer is thinner than 5 µm but an alternate way of edge protection during edge polishing, which is more reproducible, is pressing a LiNbO₃ counterpart against the substrate surface without use of epoxy. The LiNbO₃ wafers are clamped tightly by a brass jig and sufficient amount of pressure is applied to the edges to minimise the air-gap between the two wafers. First lapping is done on silicon carbide abrasive papers. This is followed by fine polishing using diamond polishing compound of particle size of 6 µm, 1 µm and 0.25 µm one after another. The near-field image of the polished output of a fabricated Ti:LiNbO₃ waveguide at 0.6328 µm wavelength for TM polarisation is shown in Fig.2.

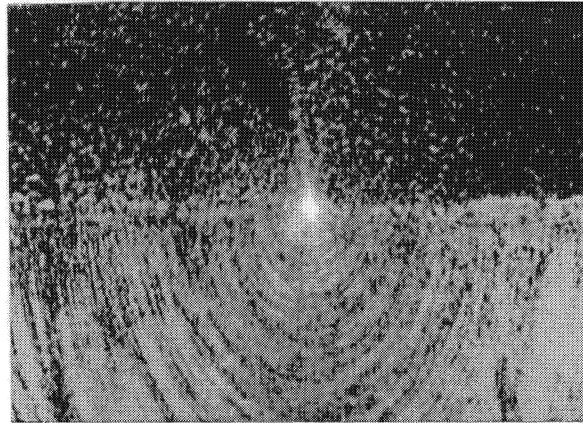


Fig.2 The near-field image of the polished output of a fabricated Ti:LiNbO₃ waveguide at 0.6328 µm wavelength for TM polarisation

Numerical Analysis

In order to get detailed information about the mode profile of waveguides, numerical analyses using the effective-index-based matrix method (EIMM) has been proposed. In this method the ray-optics approach [6] has been used to describe plane wave propagation. In the ray-optics approach, the plane waves are represented by rays drawn normal to the planes of constant phase. If the waveguide is discretised into a number of layers, then the ray incident at any layer will undergo reflection and refraction at the interface of two layer and the electric field of the i -th layer (Fig. 3) can be written as

$$\vec{E}_i = u_i^+ E_i^+ e^{j\Delta_i} \exp[j(\omega t - K_i \cos\theta_i x - \beta y)] + u_i^- E_i^- e^{-j\Delta_i} \exp[j(\omega t + K_i \cos\theta_i x - \beta y)] \quad (1)$$

where $\Delta_1 = \Delta_2 = 0$; $\Delta_3 = K_3 d_2 \cos\theta_3$, $\Delta_4 = K_4 \cos\theta_4 (d_2 + d_3)$;

$$\Delta_i = K_i \cos\theta_i (d_2 + d_3 + \dots + d_{i-1}); \quad K_i = K_0 n_i = \frac{\omega}{c} n_i;$$

$$\beta_i = K_1 \sin\theta_1 = K_2 \sin\theta_2 = \dots = K_0 n_i \sin\theta_i = \beta$$

β is the propagation constant along or parallel to the layers and is the same for all the layers to

satisfy the law of refraction. $n_i \sin\theta_i$ is called the effective refractive index. $\vec{E}_i^+$ and $\vec{E}_i^-$ are the

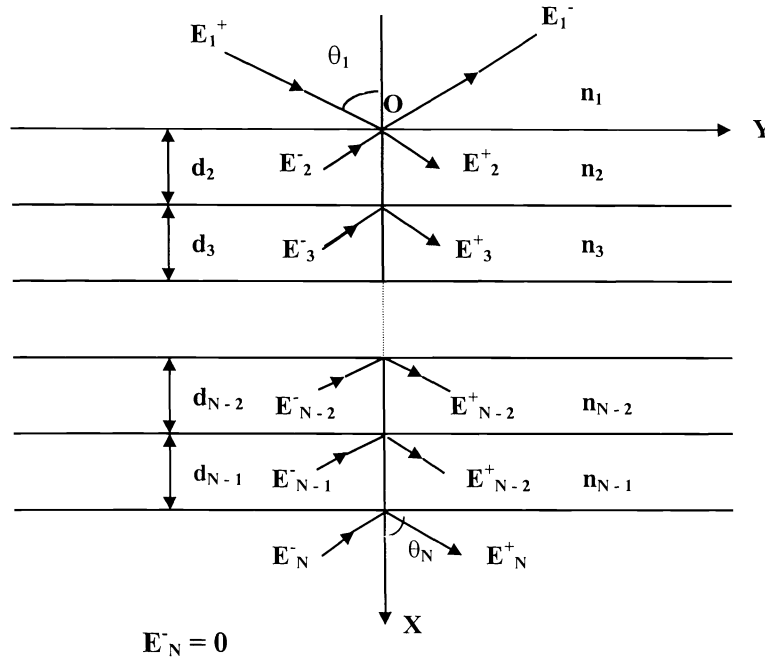


Fig. 3 Propagation of light through a multi-layered medium

magnitudes of inclined downward and upward components of the electric field vector for the i -th layer with corresponding unit vectors $\vec{u}_i^+$ and $\vec{u}_i^-$ respectively.

Now if the incident light is plane polarised at an angle α with respect to the plane of incidence, then $\vec{E}_i$ will have two components $\vec{E}_{i||}$ and $\vec{E}_{i\perp}$ parallel and perpendicular to the plane of incidence. The magnitude of the electric field vector $\vec{E}_i$ in the i -th layer may therefore be expressed as

$$E_i = (E_{i||}^2 + E_{i\perp}^2)^{1/2}$$

where $E_{i||}$ and $E_{i\perp}$ are the magnitudes of the vectors $\vec{E}_{i||}$ and $\vec{E}_{i\perp}$ respectively.

The magnitude of $\vec{E}_{i\perp}$ can be easily calculated as the unit vectors are equal ($\vec{u}_\perp^+ = \vec{u}_\perp^- = \vec{u}_\perp$), but the unit vectors for $\vec{E}_{i||}$, $\vec{u}_{i||}^+$ and $\vec{u}_{i||}^-$, have different directions separated by $2\theta_i$ and so $\vec{E}_{i||}$ is to be split into longitudinal and transverse components.

The unit vectors $\vec{u}_{i||}^+$ and $\vec{u}_{i||}^-$ may therefore be split as

$$\vec{u}_{i||}^+ = \vec{u}_l \cos \theta_i + \vec{u}_t \sin \theta_i$$

$$\vec{u}_{i||}^- = -\vec{u}_l \cos \theta_i + \vec{u}_t \sin \theta_i$$

where the suffices l and t stands for longitudinal and transverse directions respectively.

Hence $\vec{E}_{i||}$ can be expressed as

$$E_{i_{ll}} = \left(|E_{i_{ll,l}}|^2 + |E_{i_{ll,t}}|^2 \right)^{1/2}$$

where $E_{i_{ll,l}}$ and $E_{i_{ll,t}}$ are the phasors of $\vec{E}_{i_{ll,l}}$ and $\vec{E}_{i_{ll,t}}$ respectively and are given in [7]. It is to be noted here that for TE modes we have $\alpha = 90^\circ$ and for TM modes, $\alpha = 0^\circ$.

The electric field magnitude can ultimately be found out, if the magnitudes of downward and upward propagating components of the electric field are known. The magnitude of these electric fields in adjacent layers are inter-related and are given in [7,18,19]. If we consider the propagating medium to be made of N layers, then there will be no return waves from the last layer and so the upward propagating component of the electric field of last layer, $E_N^- = 0$. Using this boundary condition we can compute the magnitude of downward and upward propagating electric fields (E_i^+ and E_i^- respectively) in any i -th layer with respect to E_N^+ . After vectorial combination of the electric field components we get the magnitudes, physical orientations and relative phases of the E -vectors in all the layers for the given layered structure of known refractive indices and thicknesses.

The computation involves only multiplication of 2 X 2 matrices and hence is extremely fast, although the matrices involve complex numbers. It allows computer-simulated observation of the light field distribution as it propagates along the guiding structure and can be used for planar, step-index and graded index waveguides with arbitrary refractive index variation. However, due to certain assumptions made in this method, it does not have the accuracy of BPM and there are some inherent problems of calculating the electric field at the boundary of the waveguide. This boundary value problem can however be eliminated by using Modified Airy Function (MAF) [20]. Eventhless, this method can act as a CAD tool for calculating mode profiles of waveguiding structures.

Results

The results of the numerical calculation of mode profiles are compared with the measured values of the fabricated waveguides. The experimental setup for measuring the mode profile is shown in Fig.4. A He-Ne laser was initially used for visual alignment which is then replaced by the laser source at 1.31 μm operating wavelength. The waveguide output is magnified and the magnified intensity profile at the image plane is scanned using an InGaAs photodetector which has a 10 μm pinhole attached to it. A high resolution X-Y-Z translational stage (minimum mechanical movement of 1 μm) is used for the measurement. The magnification is calibrated accurately and special care is taken so that optical beam is incident perpendicular to the scanning plane. For a well-aligned setup, it is possible to obtain near-field measurements repeatable to a high degree of accuracy. Measured and calculated lateral and depth-wise near-field mode profiles at 1.31 μm wavelength of a 6.3 μm Ti:LiNbO₃ waveguides diffused at 1050 $^\circ\text{C}$ for 6 hrs. for both TE and TM modes are shown in Figs.5(a) and 5(b), respectively. It is already established, that under the above specified fabrication condition the waveguide will be single-moded for both TE and TM polarisations at 1.31 μm wavelength. From the measured values for both polarisations, it is seen that the lateral intensity profile of the waveguide is Gaussian in nature, whereas the depth intensity profile is Hermite-Gaussian in shape, which indicates the single-mode nature of the waveguide at both the polarisations. The numerical calculation also reveals that the propagation is single mode in nature both in lateral and depth directions. As can be seen from the figs. that the maximum modal intensity of the single-mode Ti:LiNbO₃ waveguide occurs a few microns (2-3 μm) below the surface and TM mode peak is closer to the surface than that of TE mode. The lateral profile is measured at that depth where the depth mode profile has the maximum value. The lateral mode profile at this depth is calculated and compared with the measured values of the lateral profile. The calculated FWHM mode-width and depth are 5.8 μm (4.8 μm) and 3.2 μm (2.6 μm) respectively for TE modes (TM modes) while the measured values are 5.9 μm (5.4 μm) and 4.1 μm (3.7 μm) for TE mode(TM mode) respectively. As can be seen from the figs. that there is a good match between the measured and calculated mode-depth and mode-width for both polarisation

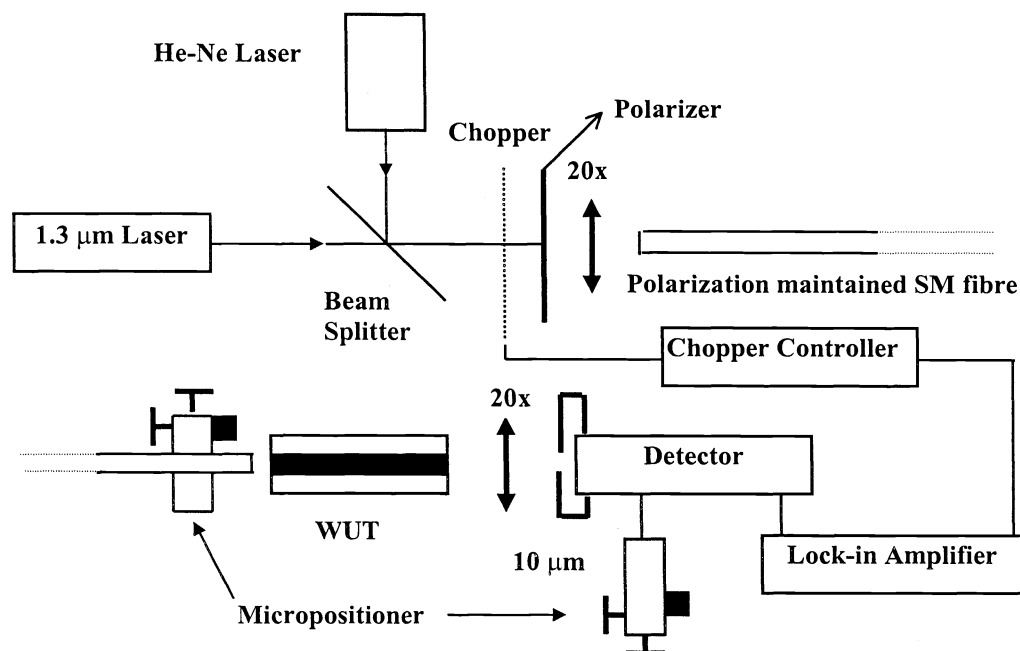


Fig.4 Setup for the experimental measurement of near-field intensity profile

indicating the validity of EIMM. TM mode is more strongly confined than TE mode because of higher refractive index change for extraordinary polarisation, resulting in lesser mode-width and mode-depth. Fig. 6 gives the lateral mode profile at the output of a directional coupler of interaction length 4.07mm. The calculated critical coupling length (L_c) of this directional coupler ($W_1=W_2 = 6.3 \mu\text{m}$, $\tau_1 = \tau_2 = 95\text{nm}$, $S = 3.1 \mu\text{m}$, $T=1050^\circ\text{C}$, $t = 6\text{hr}$) for TE mode at $1.31 \mu\text{m}$ operating wavelength is around 1.24mm. At the specified interaction length the calculated peak power ratio ($P_{\pm}/P_{\times}$) is 0.28 while the measured value is 0.3. In the fabricated directional coupler the two output ports of the directional coupler are separated out by a distance of $210 \mu\text{m}$ by S-shaped bends for butt-coupling of fibres to waveguide. In the three waveguide power divider [21], power is launched in the middle waveguide and is equally divided in the two outer waveguides after any particular distance. Only at L_c (for calculation of L_c refer to [7]) there will be 3dB power splitting of the input light. The normalised output of such a three-waveguide power divider at a particular distance is shown in Fig. 7 for TE polarisation. The figs. reveals a good match between the measured value and the calculated value.

Conclusions

A new model for calculation of mode profile of channel waveguides has been proposed. This model uses the matrix method which does not require any iteration and is thus extremely fast. The method may also be extended to calculate the field profile of periodically segmented waveguides and Y-branch splitters at any propagating distance. The computed results are verified with the measured values of the waveguides and directional couplers fabricated by the authors. Although only the fabricated Ti:LiNbO_3 waveguides are compared but the model can be used for other step and graded-index waveguides provided the refractive index profiles are available. The model can be used for any polarisation but only TE and TM modes are considered in this paper.

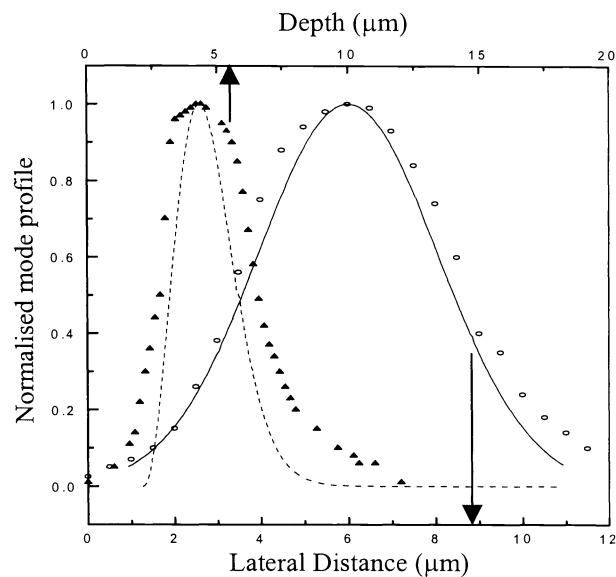
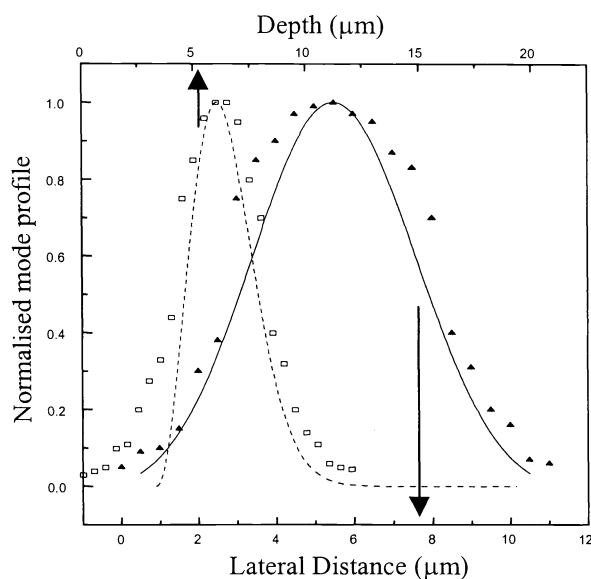


Fig 5 Calculated and measured lateral and depth mode profile of a waveguide for (a) TE (b) TM polarisation. Fabrication parameters: Titanium strip width = $6.3 \mu\text{m}$, thickness = 95 nm , diffusion time = 6 Hr (wet oxygen), Diffusion temp = 1050°C , Wavelength = $1.31 \mu\text{m}$

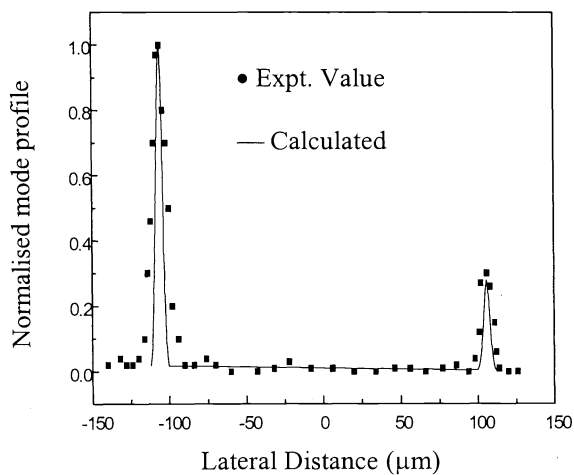


Fig 6 Calculated and measured lateral mode profile of a directional coupler at a propagating distance of 4.07 mm for TE polarisation. Fabrication parameters: Titanium strip width of each waveguide = $6.3 \mu\text{m}$, gap between each waveguide = $3.1 \mu\text{m}$, thickness of deposited titanium on each waveguide = 95 nm , diffusion time = 6 Hr (wet oxygen), Diffusion temp = 1050°C , Wavelength = $1.31 \mu\text{m}$

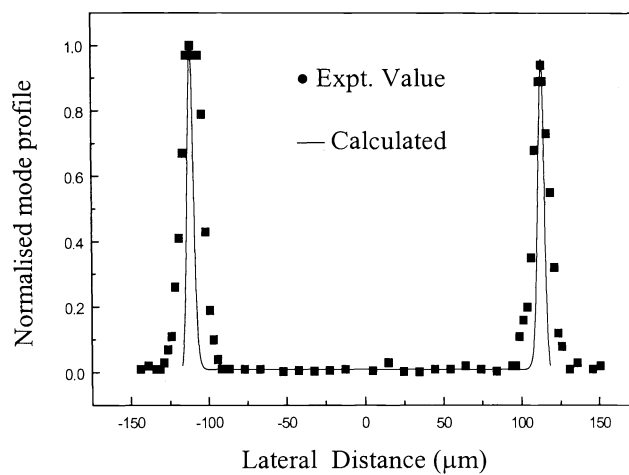


Fig 7 Calculated and measured lateral mode profile at the output of a three waveguide power divider for TE polarisation. Fabrication parameters: Titanium strip width of each waveguide = $6.3 \mu\text{m}$, gap between each waveguide = $3.1 \mu\text{m}$, thickness of deposited titanium on each waveguide = 95 nm , diffusion time = 6 Hr (wet oxygen), Diffusion temp = 1050°C , Wavelength = $1.31 \mu\text{m}$

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